

Title of Invention:**AI-DRIVEN IIoT FRAMEWORK FOR REAL-TIME INDUSTRIAL EQUIPMENT MONITORING****Abstract:**

This invention concerns an Artificial Intelligence (AI)-based Industrial Internet of Things (IIoT) system designed for the real-time monitoring, condition evaluation, anomaly detection, and predictive maintenance of industrial machinery. The system includes a number of industrial sensors which are set up to collect operating parameters such as vibration, temperature, pressure, current, voltage, rotational speed, acoustic signals and other parameters that are specific to the equipment; an IIoT edge gateway that is capable of acquiring, pre-processing, normalising and securely transmitting the sensor data; an edge intelligence module that carries out real-time feature extraction and initial anomaly detection; a communication module that sends the processed data to an industrial data platform; an AI/ML analytics engine that is able to identify abnormal operating conditions, classify the states of the equipment, estimate its health and predict possible faults; and a visualization and decision-support module that produces equipment-health indicators, alerts, maintenance recommendations and historical analytics. The framework also includes continuous feedback from the equipment's condition to the AI/ML analytics engine so that the models can be adaptively updated. The framework in question decreases the need for manual inspection, allows for the early detection of abnormalities in the equipment, supports predictive maintenance, enhances the availability of the equipment and promotes intelligent industrial decision-making.

Title of Invention:

AI-DRIVEN IIoT FRAMEWORK FOR REAL-TIME INDUSTRIAL EQUIPMENT MONITORING

Description:

Computer Science & Engineering: This invention is generally concerned with the Industrial Internet of Things (IIoT), artificial intelligence, machine learning, industrial automation, and condition monitoring and predictive maintenance. More specifically, the invention concerns an AI-driven IIoT framework which is able to acquire multi-parameter data from industrial equipment in real time, process that data at an edge or gateway level, detect abnormal operating conditions by means of machine-learning models, estimate the condition of the equipment, predict possible failures and offer practical maintenance recommendations.

Background of the Invention

Industrial equipment like motors, pumps, compressors, turbines, conveyors, generators, machine tools, and production-line equipment works under different loads, temperatures, pressures, and environmental conditions. If equipment fails without warning, it can cause production downtime, higher maintenance costs, equipment damage, and safety risks.

Standard methods of maintenance usually consist of corrective maintenance and scheduled preventive maintenance. Corrective maintenance is carried out after a failure or malfunction has occurred, while preventive maintenance is typically carried out at set time intervals. Such approaches do not always correctly reflect the true condition of health or the degree of degradation of individual equipment.

Monitoring systems which are based on sensors are able to continuously collect data on various parameters such as vibration, temperature, current, pressure and rotational speed. Yet traditional monitoring systems usually only show the measured values and produce alarms based on thresholds. They thus find it difficult to detect the complex relationships between the different sensor parameters and to identify small deviations before a serious failure takes place.

The quick advancement of IIoT allows industrial equipment to be connected via sensors, gateways, communication networks and cloud or edge computing platforms. Yet having large amounts of data from industrial sensors by itself does not result in intelligent equipment-health assessments.

An integrated framework is therefore required which combines IIoT sensing, edge processing, secure communication and AI/ML-based analytics if equipment condition is to be continuously assessed, anomalies are to be detected, potential failures are to be predicted and maintenance decisions are to be supported.

Objects of the Invention

Industrial equipment like motors, pumps, compressors, turbines, conveyors, generators, machine tools, and production-line equipment works under different loads, temperatures,

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Summary of the invention

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1. Overall Architecture

The proposed framework comprises:

1. Industrial Equipment Layer;
2. Sensor and Data Acquisition Layer;
3. IIoT Edge Gateway Layer;
4. Communication and Connectivity Layer;
5. AI/ML Analytics Layer;
6. Industrial Data Storage Layer; and
7. Visualization and Decision-Support Layer.

The industrial equipment may comprise motors, pumps, compressors, generators, turbines, conveyors, robotic equipment, CNC machines, production machinery or other industrial assets.

2. Sensor Layer

A plurality of sensors is deployed on or around the industrial equipment.

The sensor layer may include:

- vibration sensor;
- MEMS accelerometer;
- temperature sensor;
- pressure sensor;
- current sensor;
- voltage sensor;
- acoustic sensor;
- rotational-speed sensor;
- humidity sensor; and
- equipment-specific process sensors.

The sensors continuously or periodically acquire operating data.

The sensor data may be represented as:

$$\mathbf{D(t)} = \{\mathbf{V(t)}, \mathbf{T(t)}, \mathbf{P(t)}, \mathbf{I(t)}, \mathbf{R(t)}, \mathbf{A(t)}, \mathbf{S(t)}\}$$

where:

$\mathbf{V(t)}$ represents vibration data;

$\mathbf{T(t)}$ represents temperature;

$P(t)$ represents pressure;

$I(t)$ represents current;

$R(t)$ represents voltage or electrical operating information;

$A(t)$ represents acoustic information; and

$S(t)$ represents speed or rotational information.

The particular sensor combination may be selected according to the type of industrial equipment.

3. Edge Acquisition and Processing

The sensor data is supplied to an IIoT edge gateway.

The edge gateway comprises a processor, memory, communication interface and data-processing module.

The gateway performs:

- sensor-data acquisition;
- timestamp synchronization;
- signal filtering;
- noise removal;
- missing-data identification;
- normalization;
- feature extraction;
- local anomaly detection;
- data compression; and
- secure transmission.

The edge processing reduces unnecessary transmission of raw data and permits rapid response to critical equipment conditions.

4. Feature Extraction

The framework extracts statistical, temporal and frequency-domain characteristics from the acquired signals.

For vibration signals, extracted features may include:

- root mean square;
- peak value;
- crest factor;
- kurtosis;
- variance;
- standard deviation;
- frequency components;

- spectral energy; and
- dominant frequency.

For temperature and electrical signals, the framework may extract:

- mean;
- maximum;
- minimum;
- rate of change;
- moving average;
- fluctuation;
- load-related characteristics; and
- temporal trends.

The extracted features form a feature vector:

$$\mathbf{F}(t) = [\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \dots, \mathbf{f}_n]$$

where each feature represents a characteristic of the equipment condition.

5. AI/ML Analytics Engine

The feature vector is supplied to the AI/ML analytics engine.

The analytics engine may include one or more models selected according to the monitoring objective.

The models may include:

- supervised classification models;
- regression models;
- decision trees;
- random forests;
- support vector machines;
- artificial neural networks;
- deep neural networks;
- autoencoders;
- clustering algorithms;
- time-series models; or
- ensemble learning models.

For an industrial asset having sufficient historical fault-labelled data, supervised learning may be used for fault classification.

Where labelled fault data is limited, unsupervised or semi-supervised anomaly detection may be employed.

6. Equipment Health Index

The framework generates a health index representing the estimated operating condition of the industrial equipment.

In one embodiment, the health index is calculated from weighted sensor features and AI/ML model outputs.

The health index may be represented by:

$$\mathbf{HI} = \mathbf{f}(\mathbf{F}, \mathbf{M}, \mathbf{H})$$

where:

HI = equipment health index;

F = extracted sensor features;

M = machine-learning model output; and

H = historical operating information.

The health index may be mapped to different equipment-condition categories such as:

- normal;
- warning;
- abnormal;
- critical.

The thresholds may be configured according to equipment type and operating conditions.

7. Anomaly Detection

The AI/ML engine continuously compares incoming equipment data with learned operating behaviour.

An anomaly score may be generated as:

$$\mathbf{AS}(\mathbf{t}) = \mathbf{d}(\mathbf{F}(\mathbf{t}), \mathbf{F}_{\text{baseline}})$$

where:

AS(t) is the anomaly score and Fbaseline represents a learned baseline feature profile.

When the anomaly score exceeds a predetermined or dynamically learned limit, an abnormal condition is identified.

The system may distinguish between transient abnormalities and persistent abnormalities by analysing the abnormality over multiple time intervals.

8. Fault Prediction

The framework may use temporal equipment data to estimate the likelihood of future failure.

Potential fault classes may include:

- bearing degradation;
- motor overheating;
- excessive vibration;
- shaft misalignment;
- electrical overload;
- lubrication-related abnormality;
- pressure abnormality;
- imbalance; and
- other equipment-specific faults.

The prediction module may produce a fault probability or predicted degradation level.

9. Decision-Support Module

The decision-support module receives outputs from the AI/ML analytics engine.

The module generates:

- real-time equipment status;
- health index;
- anomaly score;
- fault category;
- severity level;
- maintenance alert;
- maintenance recommendation; and
- historical trend.

The information may be displayed through a dashboard accessible through an industrial workstation, computer, tablet or mobile interface.

10. Alert Mechanism

The framework may classify alerts into different severity levels.

For example:

Level 0 – Normal: Equipment operates within learned normal conditions.

Level 1 – Warning: Minor deviation detected.

Level 2 – Abnormal: Persistent deviation or probable fault detected.

Level 3 – Critical: High-probability failure or dangerous operating condition detected.

The alert may be transmitted to an operator, maintenance engineer or industrial control system.

11. Secure Communication

The communication layer provides connectivity between sensors, edge gateways and the analytics platform.

Communication may employ wired or wireless industrial communication technologies.

The framework may incorporate authentication, encrypted communication, access control and integrity verification to reduce unauthorized access to industrial monitoring data.

12. Feedback and Adaptive Learning

The framework may record subsequent equipment conditions and maintenance events.

For example, when a predicted fault is confirmed during inspection or maintenance, the corresponding event is associated with the sensor data that preceded the fault.

The resulting information may be used to retrain or update the AI/ML model.

Thus, the framework may progressively adapt to the individual operating characteristics of an industrial asset.

Operation of the invention

The operation of the proposed system comprises the following steps:

1. Deploying a plurality of sensors on industrial equipment.
2. Acquiring real-time equipment parameters.
3. Transmitting the sensor information to an IIoT edge gateway.
4. Filtering and preprocessing the acquired data.
5. Extracting relevant time-domain, statistical and frequency-domain features.
6. Generating a feature vector representing the equipment condition.
7. Supplying the feature vector to an AI/ML analytics engine.
8. Comparing the current equipment condition with a learned baseline.
9. Calculating an anomaly score and equipment-health index.
10. Identifying and classifying abnormal operating conditions.
11. Predicting potential equipment faults.
12. Generating a maintenance alert or recommendation.
13. Displaying equipment condition and analytics through a dashboard.
14. Recording actual maintenance outcomes.
15. Updating the AI/ML model based on subsequent equipment behaviour.

Advantages of the invention

The proposed invention provides several technical and operational advantages, including:

1. Real-time monitoring of industrial equipment.
2. Continuous multi-parameter data acquisition.
3. Edge-based preprocessing and rapid anomaly detection.
4. AI/ML-based intelligent condition assessment.

5. Early detection of abnormal equipment behaviour.
6. Predictive rather than solely reactive maintenance.
7. Reduced dependence on manual inspection.
8. Scalable monitoring of multiple industrial assets.
9. Equipment-specific health assessment.
10. Historical trend analysis and maintenance support.
11. Reduced unnecessary transmission of raw sensor data.
12. Adaptive improvement through feedback-based learning.
13. Improved equipment availability and operational reliability.

Industrial applicability

The invention is applicable to manufacturing plants, process industries, power-generation facilities, automotive industries, chemical plants, pharmaceutical manufacturing, food-processing industries, mining operations, logistics facilities, robotics-based production systems and other industrial environments where continuous equipment monitoring is required.

Title of Invention:

AI-DRIVEN IIoT FRAMEWORK FOR REAL-TIME INDUSTRIAL EQUIPMENT MONITORING

Drawings:

FIG. 1: Overall architecture of the AI-driven IIoT framework.

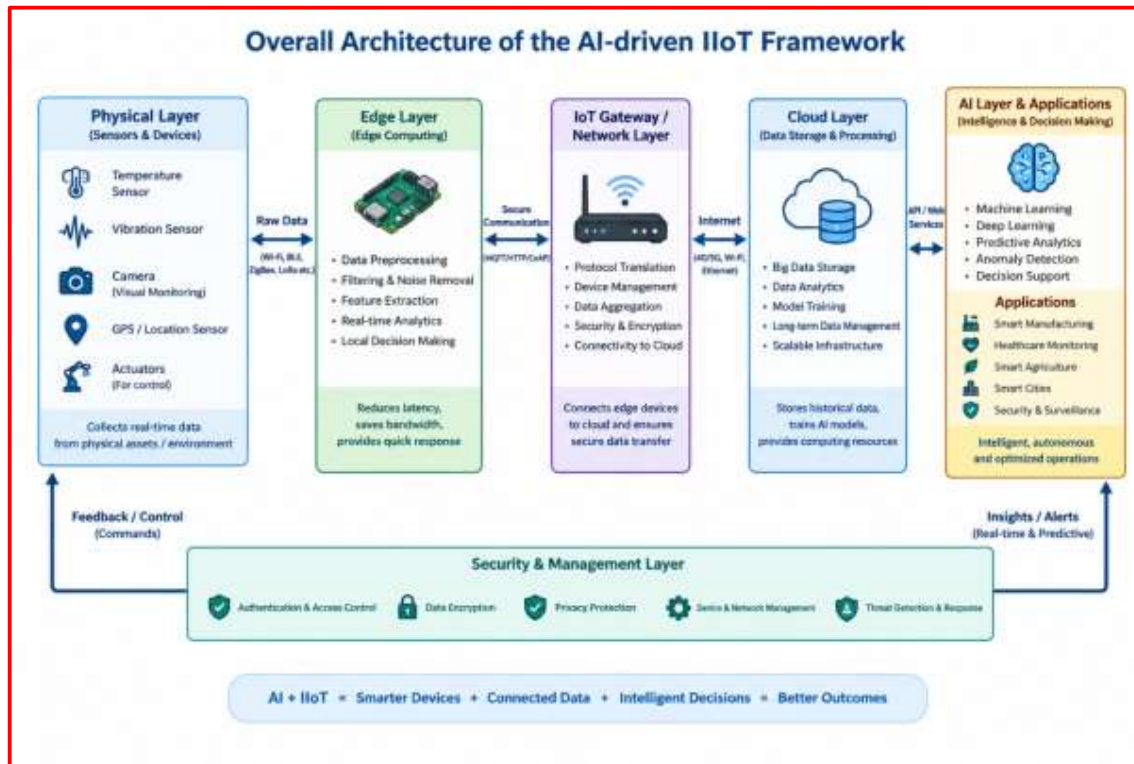


FIG. 2: Industrial sensor and data-acquisition subsystem.

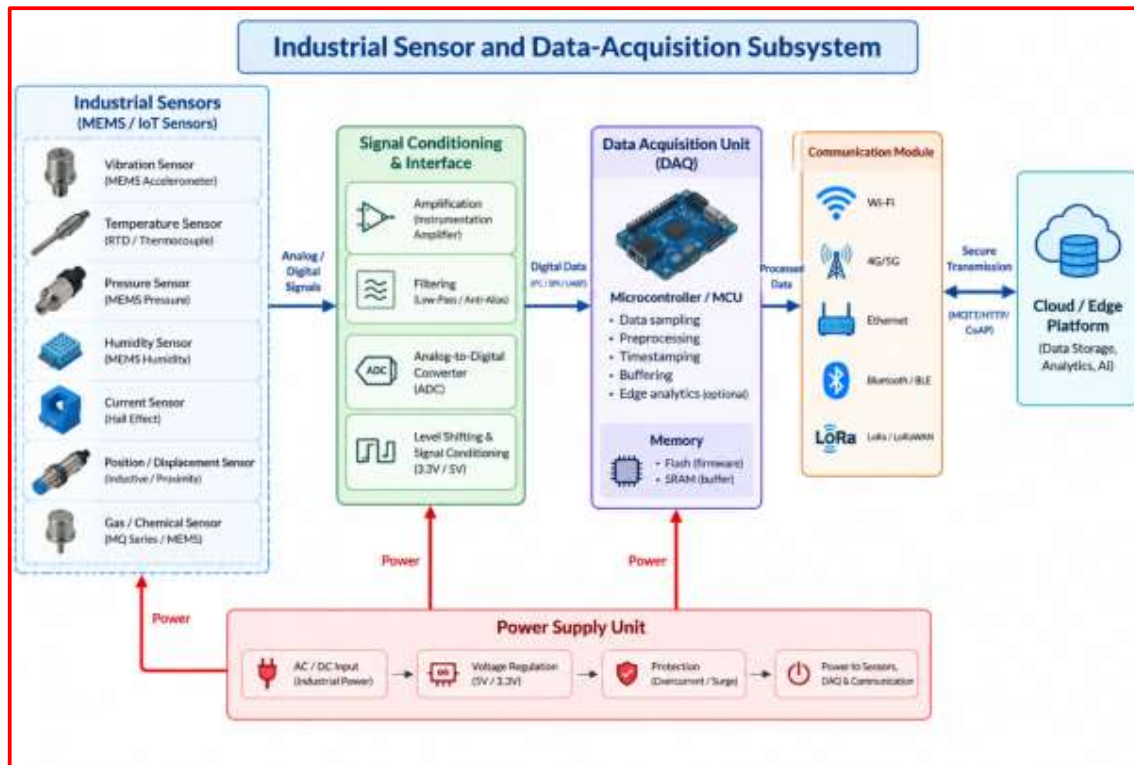


FIG. 3: Architecture of the IIoT edge gateway.

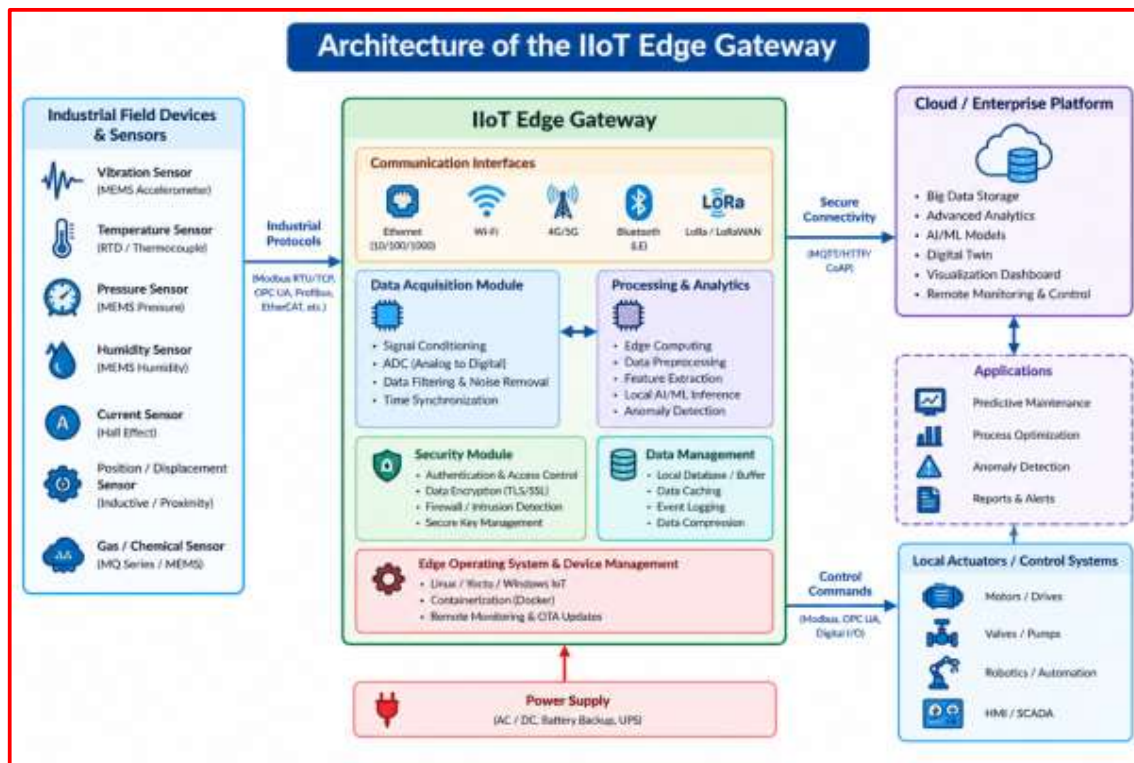


FIG. 4: AI/ML-based anomaly detection and equipment-health assessment process.

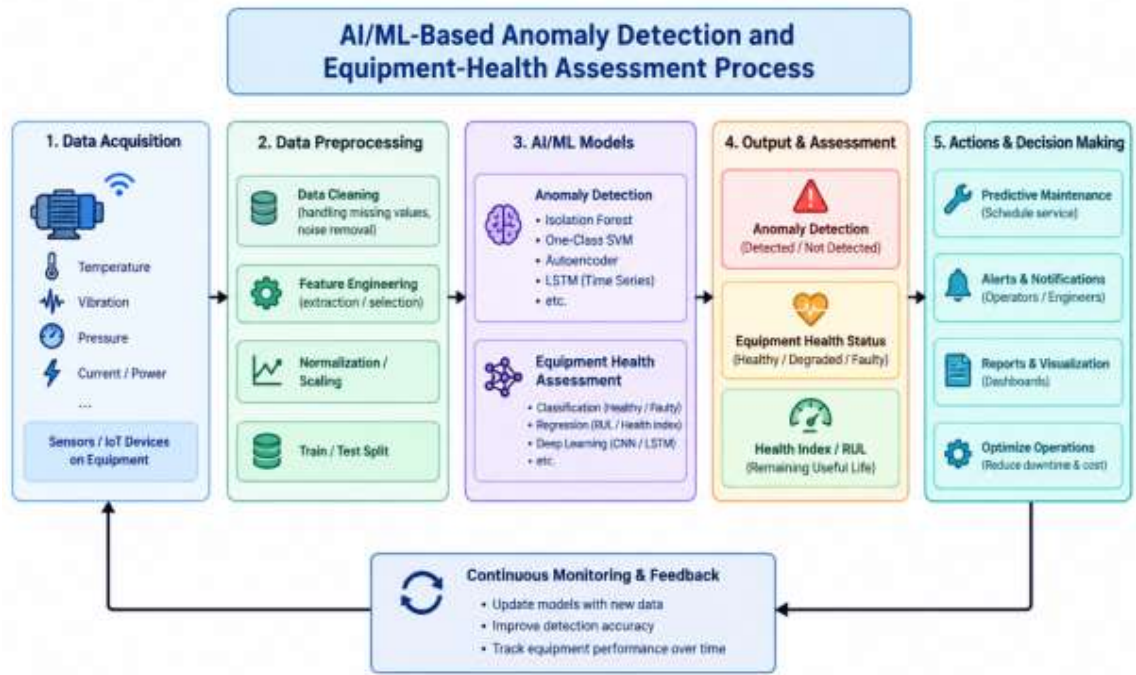


FIG. 5: Predictive maintenance and fault-prediction workflow.

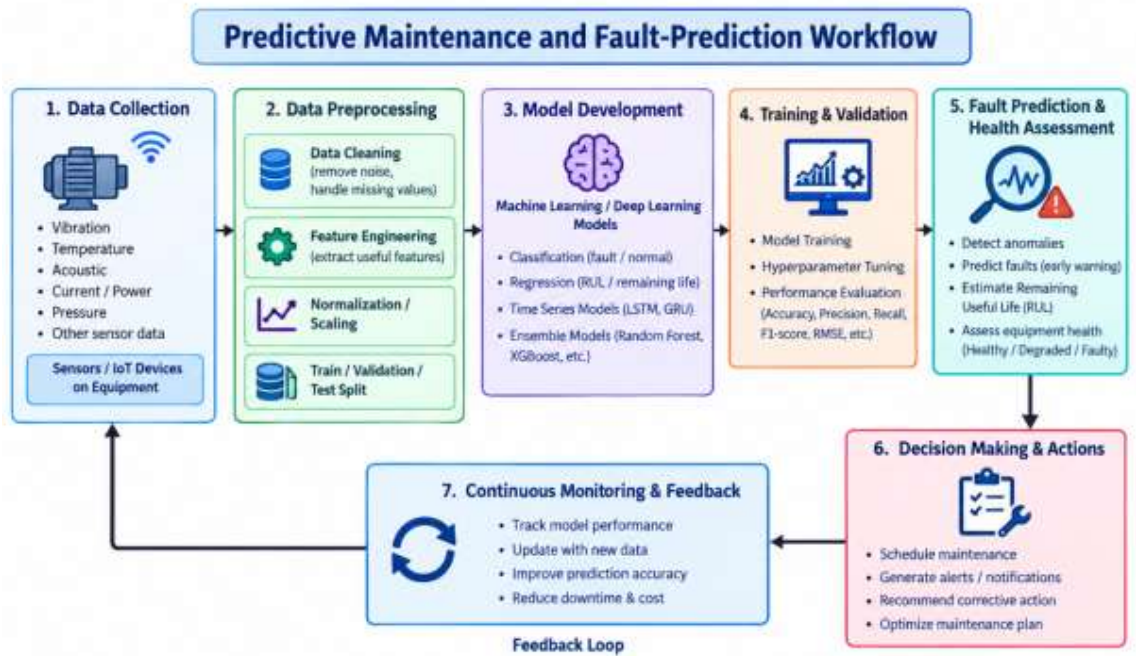
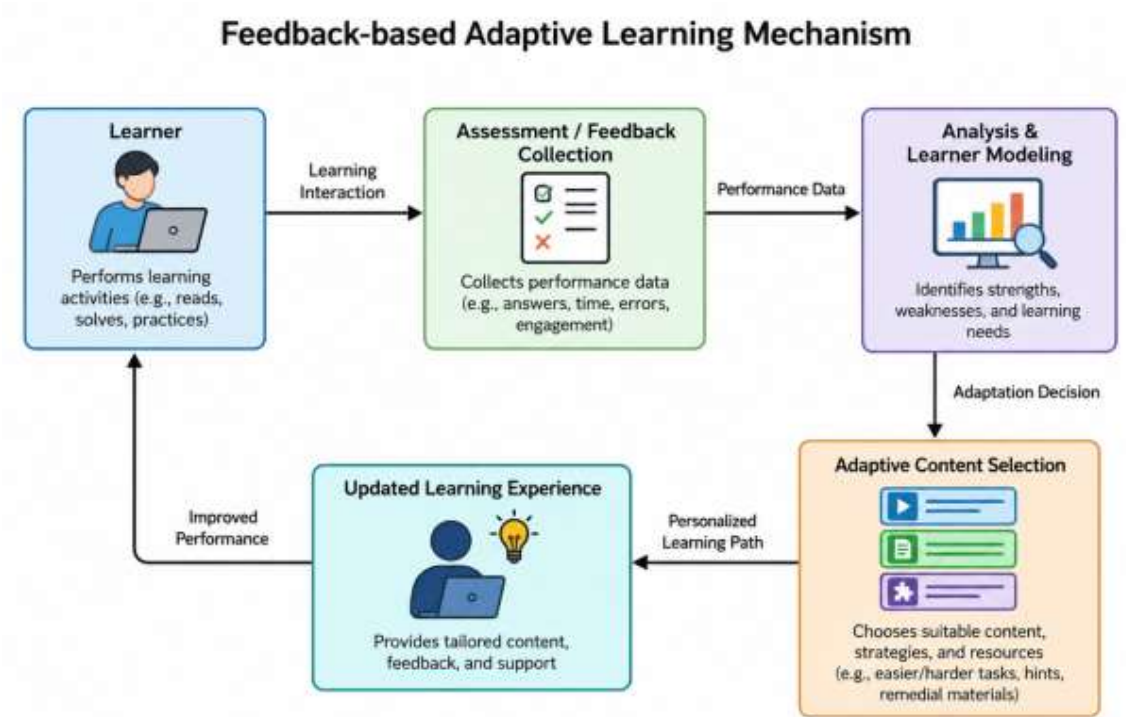


FIG. 6: Industrial dashboard showing equipment health, anomaly score and alerts.



FIG. 7: Feedback-based adaptive learning mechanism.



Explanation of Figures

FIG. 1 – Overall Architecture of the AI-Driven IIoT Framework

1. FIG. 1 illustrates the overall architecture of the proposed AI-driven Industrial Internet of Things (IIoT) framework for real-time industrial equipment monitoring.
2. The architecture consists of the industrial equipment layer, sensor and data-acquisition layer, IIoT edge gateway, communication layer, AI/ML analytics layer, data-storage layer, and visualization and decision-support layer.
3. Sensors acquire parameters such as vibration, temperature, pressure, current, voltage, acoustic signals, and rotational speed from industrial equipment.
4. The edge gateway preprocesses the acquired data and transmits relevant information to the AI/ML analytics engine through a communication module.
5. The AI/ML engine analyses the processed data to assess equipment health, detect abnormalities, predict faults, and provide maintenance-related information through the decision-support module.

FIG. 2 – Industrial Sensor and Data-Acquisition Subsystem

1. FIG. 2 illustrates the sensor and data-acquisition subsystem used for collecting real-time operating parameters from industrial equipment.
2. A plurality of sensors may include vibration sensors, MEMS accelerometers, temperature sensors, pressure sensors, current and voltage sensors, acoustic sensors, and rotational-speed sensors.
3. The sensors continuously or periodically acquire equipment operating data according to the requirements of the particular industrial asset.
4. The acquired data may include vibration, temperature, pressure, electrical, acoustic, and rotational-speed information represented as a multi-parameter data set.
5. The subsystem provides the collected sensor information to the IIoT edge gateway for filtering, normalization, feature extraction, anomaly detection, and subsequent AI/ML analysis.

FIG. 3 – Architecture of the IIoT Edge Gateway

1. FIG. 3 illustrates the architecture of the IIoT edge gateway responsible for acquiring and locally processing sensor data.
2. The gateway comprises processing capability, memory, communication interfaces, and data-processing functionality for handling incoming sensor information.
3. The gateway performs operations including timestamp synchronization, signal filtering, noise removal, missing-data identification, normalization, and feature extraction.
4. Local anomaly detection and data compression may also be performed at the gateway before transmission to the remote analytics platform.
5. Edge processing reduces unnecessary transmission of raw sensor data and enables faster responses to critical equipment conditions.

FIG. 4 – AI/ML-Based Anomaly Detection and Equipment-Health Assessment Process

1. FIG. 4 illustrates the AI/ML-based process for detecting abnormal equipment behaviour and assessing the health condition of industrial equipment.
2. Processed sensor signals are converted into statistical, temporal, and frequency-domain features forming a feature vector.
3. The feature vector is supplied to an AI/ML analytics engine, which may employ classification, regression, neural-network, tree-based, clustering, or other machine-learning models.
4. The system generates an anomaly score by comparing the current feature profile with a learned baseline and calculates an equipment health index using sensor features, model outputs, and historical information.
5. The equipment condition can be classified into normal, warning, abnormal, or critical states to support timely identification of degradation and abnormal operation.

FIG. 5 – Predictive Maintenance and Fault-Prediction Workflow

1. FIG. 5 illustrates the predictive maintenance and fault-prediction workflow of the proposed framework.
2. The system analyses temporal sensor characteristics and historical equipment information to estimate the likelihood of future equipment failure.
3. Potential faults may include bearing degradation, motor overheating, excessive vibration, shaft misalignment, electrical overload, lubrication abnormalities, pressure abnormalities, and imbalance.
4. The prediction module may generate a fault probability or predicted degradation level for the monitored equipment.
5. Based on the predicted fault and its severity, the decision-support module generates maintenance alerts and recommendations, thereby supporting predictive rather than solely reactive maintenance.

FIG. 6 – Industrial Dashboard Showing Equipment Health, Anomaly Score and Alerts

1. FIG. 6 illustrates the industrial dashboard used to present equipment-health information, anomaly scores, alerts, and related analytics.
2. The dashboard can display the real-time equipment status, equipment health index, anomaly score, fault category, and severity level.
3. It can provide maintenance alerts and recommendations to operators and maintenance personnel.
4. Historical trends can be displayed to assist in analysing equipment behaviour and supporting maintenance decisions.
5. The dashboard may be accessed through an industrial workstation, computer, tablet, or mobile interface, enabling practical monitoring and decision support.

FIG. 7 – Feedback-Based Adaptive Learning Mechanism

1. FIG. 7 illustrates the feedback-based adaptive learning mechanism through which actual equipment conditions and maintenance outcomes are fed back to the AI/ML analytics engine.
2. The system records subsequent equipment conditions, inspection results, maintenance events, and actual failure outcomes.

3. When a predicted fault is confirmed during inspection or maintenance, the corresponding event is associated with the sensor data that preceded the fault.
4. The resulting information is supplied to the AI/ML model for updating or retraining the equipment-monitoring model.
5. This feedback mechanism enables the monitoring framework to progressively adapt to the individual operating characteristics of an industrial asset and improve its future monitoring and prediction capability.

Title of Invention:

AI-DRIVEN IIoT FRAMEWORK FOR REAL-TIME INDUSTRIAL EQUIPMENT MONITORING

Claims:

Claim 1

An **AI-driven Industrial Internet of Things (IIoT) framework for real-time industrial equipment monitoring**, comprising a plurality of sensors configured to acquire operating parameters of industrial equipment; an IIoT edge gateway configured to receive and preprocess sensor data; a communication module configured to transmit processed data; an artificial intelligence and machine-learning analytics engine configured to analyse the processed data; and a decision-support module configured to generate an equipment-health condition, detect abnormal operating conditions and provide an indication of potential equipment faults.

Claim 2

The framework as claimed in claim 1, wherein the plurality of sensors comprises at least one of a vibration sensor, MEMS accelerometer, temperature sensor, pressure sensor, current sensor, voltage sensor, acoustic sensor and rotational-speed sensor.

Claim 3

The framework as claimed in claim 1, wherein the IIoT edge gateway is configured to perform at least one of noise filtering, signal conditioning, normalization, timestamp synchronization, missing-data identification, feature extraction, local anomaly detection and data compression before transmitting sensor information to the AI/ML analytics engine.

Claim 4

The framework as claimed in claim 1, wherein the AI/ML analytics engine is configured to generate an anomaly score and an equipment-health index by analysing extracted features from real-time sensor data in comparison with a learned baseline operating profile of the industrial equipment.

Claim 5

The framework as claimed in claim 1, wherein the AI/ML analytics engine is configured to classify equipment operating conditions into at least normal, warning, abnormal and critical conditions and generate a corresponding alert based on the classified condition.

Claim 6

The framework as claimed in claim 1, wherein the AI/ML analytics engine is further configured to predict a potential equipment fault based on temporal sensor characteristics and historical equipment information and wherein the decision-support module generates a maintenance recommendation based on the predicted fault.

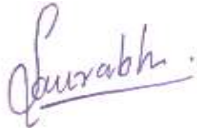
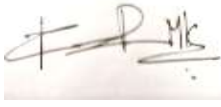
Claim 7

The framework as claimed in claim 1, further comprising a feedback mechanism configured to associate actual equipment maintenance or failure outcomes with corresponding historical sensor data and provide the associated information to the AI/ML analytics engine for updating or retraining an equipment-monitoring model.

Title of Invention:**AI-DRIVEN IIoT FRAMEWORK FOR REAL-TIME INDUSTRIAL EQUIPMENT MONITORING****Conclusion**

The proposed invention provides an AI-driven Industrial Internet of Things (IIoT) framework for real-time monitoring, condition assessment, anomaly detection, fault prediction, and predictive maintenance of industrial equipment. The framework integrates multi-parameter industrial sensing, edge-based data acquisition and preprocessing, secure communication, AI/ML-based analytics, equipment-health assessment, predictive fault analysis, and visualization and decision-support capabilities within a unified architecture. The invention enables continuous acquisition and analysis of parameters such as vibration, temperature, pressure, current, voltage, acoustic signals, and rotational speed. By performing preprocessing and feature extraction at the edge gateway, the framework can reduce unnecessary transmission of raw sensor data and support rapid detection of abnormal equipment conditions. The AI/ML analytics engine further enables equipment-health estimation, anomaly scoring, condition classification, and prediction of potential equipment faults based on real-time and historical operating information. A significant aspect of the invention is the feedback-based adaptive learning mechanism. Actual maintenance and failure outcomes are associated with the corresponding historical sensor data and used to update or retrain the AI/ML monitoring model. Consequently, the framework can progressively adapt to the operating characteristics of individual industrial assets and improve its monitoring and predictive capabilities over time. The proposed framework reduces dependence on manual inspection and conventional threshold-based monitoring while supporting early identification of abnormal behaviour and predictive maintenance. It can improve equipment availability, operational reliability, maintenance planning, and intelligent industrial decision-making. The framework is applicable to manufacturing plants, power-generation facilities, automotive industries, chemical and pharmaceutical plants, food-processing industries, mining, logistics, robotics-based production systems, and other industrial environments requiring continuous equipment monitoring. Accordingly, the invention provides an integrated and adaptable technical framework for transforming conventional industrial equipment monitoring into an intelligent, data-driven, real-time, and feedback-enabled predictive maintenance system.

Title of Invention:**AI-DRIVEN IIoT FRAMEWORK FOR REAL-TIME INDUSTRIAL EQUIPMENT MONITORING****Applicant(s) & Inventor (s) details:**

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